Is the NFL Combine Indicative of Player Performance in the NFL?

IST 707 — Machine Learning | Midterm Project Report
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Project overview

NFL teams invest substantial draft capital and money in players based on a combination of college performance, film, medical evaluations, interviews, and athletic testing. This project examines whether the NFL Scouting Combine predicts long-term NFL performance, whether the relationship differs by position, and whether athletic tests add information beyond draft selection. We studied three complementary approaches: decision trees and random forests predicting career value, K-means athlete profiles followed by classification of multi-year starters, and tuned XGBoost regression predicting career value. Across the three approaches, we found that **Combine performance has some relationship with NFL outcomes, but draft position is a much stronger predictor.** The best reported XGBoost model reached approximately **0.33 held-out R^2** with draft information, compared with **0.08** using Combine features alone. For the separate question of whether a player starts three or more seasons, Combine-only classification reached approximately **0.68 ROC AUC**, improving to **0.87** when draft position was included. Based on these results, we think Combine tests are most useful alongside film, college production, and other scouting information.

1. Problem, research questions, and definition of success

Every spring, NFL prospects complete a series of standardized athletic measurements and drills. Teams may use these results to help compare players, but a fast sprint or impressive vertical jump does not automatically translate to a productive professional career. Our project asks: **(1)** How well do Combine measurements predict NFL career value? **(2)** Does their usefulness depend on position? **(3)** Are there meaningful athletic profiles or performance thresholds that help identify future NFL contributors? **(4)** Does Combine information improve predictions after accounting for where a player was drafted?

We define **Statistical success** as performance above a defensible null baseline on unseen data, using R^2 and error measures for regression and ROC AUC for starter classification. A null regression model predicting the training-set average should produce approximately zero test R^2 when evaluated on a comparable test distribution; a random-ranking classifier has expected AUC of 0.50. These are reference benchmarks, not evidence that a particular random experiment was executed. We define **Added value** more strictly: Combine variables must improve held-out performance over a draft-only model by a meaningful and repeatable amount. Finally, **Practical success** means the model could help identify overlooked prospects, challenge overreactions to a single workout, or improve position-specific scouting decisions. Predictive correlation alone does not prove a drill causes career success.

Our results do better than the baseline, especially when predicting multi-year starters, but **the Combine has not yet shown meaningful added value for predicting career value once draft position is included**: draft-only and draft-plus-Combine XGBoost both produced R^2 near 0.33. We have not demonstrated that using the models improves real draft decisions or financial outcomes.

2. Stakeholders and why the findings matter

General managers and front offices decide how to allocate scarce draft picks and roster resources. The most relevant question for them is not whether a drill correlates with career outcomes, but whether it helps find players whose future value is greater than their expected value at a given draft slot. **Scouts and analytics departments** can use position-specific evidence to decide how much weight to place on a 40-yard dash, agility test, or jump alongside film, production, and medical information. **Owners and team executives** benefit when evidence prevents costly decisions driven by attention-grabbing workouts rather than demonstrated predictive value. **Players, agents, and trainers** can use the findings to prioritize preparation, while recognizing that athletic testing does not capture all aspects of football ability.

For example, our position-specific results suggest that speed or agility may provide a modest signal for running backs, offensive linemen, tight ends, and linebackers. That does not justify imposing the same 40-yard standard across positions. Equally, a draft-only model performing well should not be mistaken for a recommendation to draft players solely based on projected draft slot: the slot summarizes decisions already made by teams and cannot itself tell a team whom to select before the draft.

3. Related work

Prior research has raised similar questions about the usefulness of pre-draft athletic testing. Kuzmits and Adams (2008) reported that many Combine drills had weak relationships with subsequent NFL performance.[1] Robbins (2010) examined the relationship between Combine measures and player outcomes and found that certain athletic measures, including speed and jumping, can have predictive relevance.[2] Teramoto and colleagues (2016) considered the role of Combine testing in performance prediction and emphasized differences across player roles.[3] These studies motivate our use of position groups and our comparison of athletic-only models against draft-informed models. We build on this work by comparing three modeling strategies, including a classification task that does not depend on a continuous career-value target.

4. Dataset and preprocessing

We used **nflverse NFL Combine and draft data** as the source, with player records linked using Pro Football Reference identifiers. The main career-value regression sample includes **5,098 drafted players from 2000–2025**, grouped into **10 position categories**. The initial Combine inputs are height, weight, 40-yard dash, bench press, three-cone drill, shuttle, broad jump, and vertical jump. The regression outcome is Pro Football Reference **weighted career approximate value (`w_av`)**, transformed into a percentile within each draft class. This percentile adjustment partially addresses the problem that recent draft classes have had fewer seasons to accumulate value, although it does not fully solve unequal career follow-up.

The starter-classification analysis uses a different sample: **5,263 players from 2000–2019 across nine position groups**, including undrafted players where outcome data were available. Its binary target is whether a player started at least three NFL seasons; approximately **29%** met that threshold. The earlier end date allows more time to observe whether a player established a multi-year career. Models 1 and 3 focus on career-value regression, while Model 2 focuses on becoming a multi-year starter. **Their results are not directly interchangeable** because the samples, outcomes, and evaluation designs differ.

Preparation included reconciling position labels, removing career statistics that would leak the target, and accounting for skipped drills. In Model 1, missing drills were left blank where supported; one-drill analyses used participants with recorded results. In Model 2, skipped tests were flagged and values were filled using

position averages before standardizing features within positions. Missingness is substantive: roughly 40% skipped the three-cone or shuttle and about 36% skipped the bench press. Players who skip a drill may differ systematically from those who complete it, which can bias apparent relationships. We consider the data credible for exploratory modeling because of the established NFL statistical sources and identifier-based matching, but matching errors, missing outcomes, and uneven follow-up remain limitations.

5. Exploratory data analysis: five statistical views

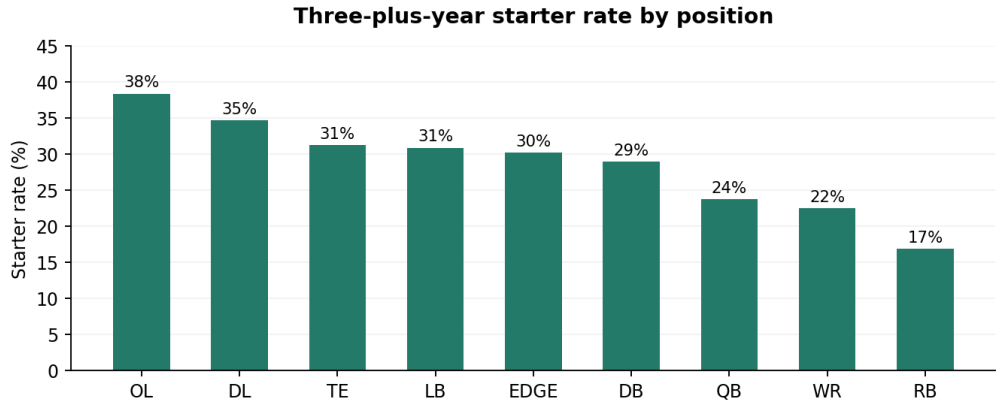
The figures below summarize our reported results. The charts were recreated from aggregate results rather than recalculated from individual player records.

Analysis 1: Career-value distribution by position

Our exploratory boxplots showed broad distributions of draft-class career-value percentiles across the position groups. The key observation is substantial overlap: no position has a uniformly high career-value outcome. The wide within-position variation reinforces why we should evaluate player-level predictive models instead of relying on a player's position alone. Our correlation analysis also showed a correlation heatmap of athletic measurements with class-adjusted value, with generally modest associations. Those correlations are descriptive, not proof of independent predictive effects.

Analysis 2: Starter rates by position

The percentage of players becoming three-plus-year starters differs noticeably across positions. This matters because a single pooled classification threshold may not mean the same thing for an offensive lineman and a running back.



Reported multi-year starter rates by position

Figure 1. Percentage of players becoming multi-year starters, by position.

Offensive linemen have the highest reported starter rate (38.4%), while running backs have the lowest (16.9%). These differences may reflect career length, roster usage, injury exposure, or position-specific opportunities; our analysis does not establish which mechanism dominates. The differences also reinforce the decision to standardize Combine performance within position.

Analysis 3: Starter rates by draft round

Our draft-round analysis showed that early-round picks become multi-year starters substantially more often than late-round picks, with a reported association of approximately $\chi^2 = 456$, $p < 0.001$. The first-round starter

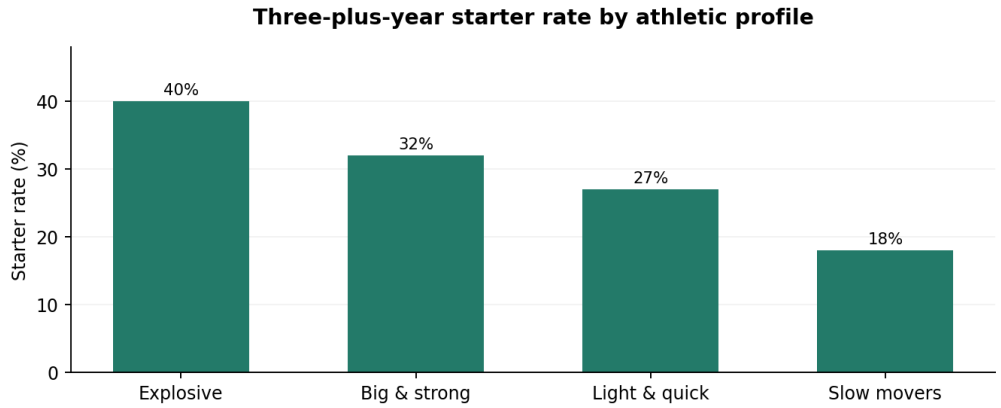
rate is visibly much higher than that for seventh-round picks. This matters because draft position reflects teams' combined judgments about talent, production, health, and expected opportunity. It creates a demanding comparison baseline: a Combine model must add information beyond what draft order already captures. This association should not be interpreted as evidence that the draft round itself causes ability.

Analysis 4: Forty-yard dash times by position

Our 40-yard dash comparisons showed clear position differences in 40-yard dash performance, and our analysis found a strong between-position difference (ANOVA $p < 0.001$). Wide receivers and defensive backs tend to run faster than larger line players. This is an important measurement lesson: a raw 4.7-second time means something different for a tight end than for a cornerback. Position-adjusted values and position-specific models are therefore more defensible than a single league-wide speed cutoff.

Analysis 5: Athletic archetypes and starter outcomes

K-means clustering identified four broad Combine profiles without using NFL outcomes in the clustering step. Their subsequent starter rates were notably different.



Starter rates by athletic profile

Figure 2. Multi-year starter rates across four athletic profiles.

The **explosive** group (962 players) had a **40%** multi-year starter rate, compared with **18%** among **slow movers** (878 players), approximately **2.2 times as high**. Our results show $\chi^2 = 119$, $p < 0.001$ for differences among groups. However, the explosive group was also drafted much earlier on average (pick 90 versus pick 145) and at a higher rate (89% versus 58%). Consequently, the raw starter-rate gap cannot be attributed entirely to athleticism. The reported silhouette score of **0.15** also indicates that the groups overlap substantially; they are descriptive profiles, not sharply separated biological types.

6. Null baselines and how models were evaluated

For **career-value regression**, our regression results use a null-baseline R^2 of approximately **0.000**. A model that predicts the average training outcome for everyone offers little individualized information; positive held-out R^2 indicates improvement over this reference. The reported Model 1 RMSE values are in **career-value percentile points**, making the errors interpretable on the outcome scale. For **starter classification**, random ranking has expected **ROC AUC = 0.50**. The 29% starter prevalence is relevant when assessing practical classification decisions; predicting 'not a starter' for every player could achieve about 71% accuracy without

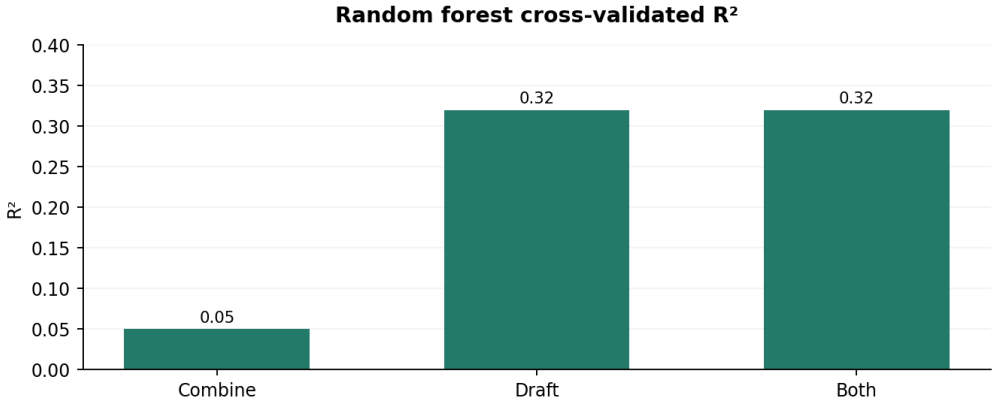
identifying a single future starter. This is why ROC AUC, recall, and ideally precision-recall metrics matter more than accuracy alone.

One important difference is that Model 1 reports both cross-validation and held-out test results, Model 2 uses a future-class split (training 2000–2015, testing 2016–2019), and Model 3 uses a tuned five-fold cross-validation process with a held-out test set. Their published scores are informative but should not be treated as a perfectly controlled head-to-head experiment.

7. Model 1: Decision tree and random forest

The first modeling approach predicts draft-class career-value percentile. An unrestricted decision tree fit the training sample perfectly (**train R^2 = 1.00**) but performed extremely poorly on held-out players (**test R^2 = −0.76; RMSE = 38.6**). This shows the problem with overfitting: the tree performed well on the training players but not on new ones. A depth-limited, pruned tree improved to **test R^2 = 0.02; RMSE = 28.8**. A random forest of 300 trees using Combine features reached **train R^2 = 0.14, CV R^2 = 0.05, test R^2 = 0.08, RMSE = 28.0**.

The strongest Model 1 result came from draft information. Draft-only forest predictions achieved **CV R^2 = 0.32, test R^2 = 0.32, RMSE = 24.0**; combining draft and Combine inputs produced **CV R^2 = 0.32, test R^2 = 0.32, RMSE = 23.9**. The 0.1-point RMSE difference is small, and rounded R^2 shows no gain. Tuning each input set separately also corrected an earlier unfair comparison in which the draft-only model overfit and drills appeared to add 3–7 percentage points of R^2. The revised estimated incremental gain was no more than three points for any position.



Random forest CV results

Figure 3. Random forest five-fold cross-validated R^2 for the model comparison.

The position-specific drill results suggest small signals for running back three-cone performance (roughly R^2 0.06), offensive-line 40 time (about 0.05), and tight end and linebacker 40 time (about 0.04). Proposed cutoffs near **7.1 seconds for RB three-cone** and **4.70 seconds for TE 40** should be treated as exploratory, not validated thresholds.

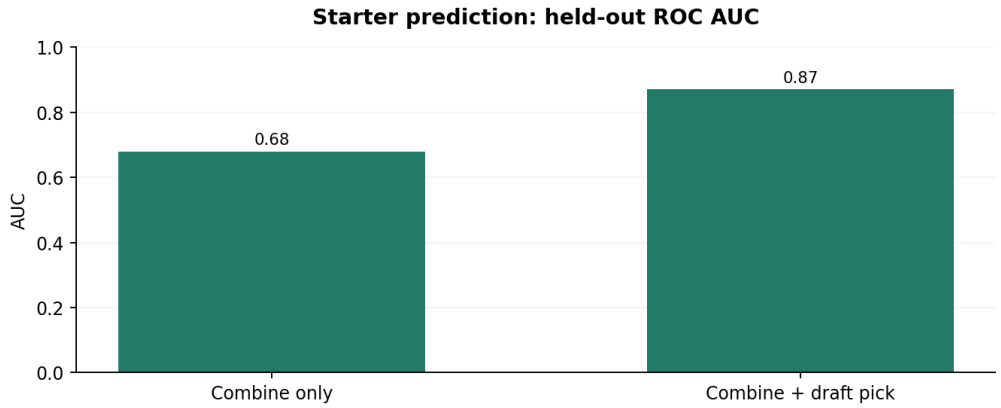
8. Model 2: K-means athletic profiles and starter classification

The second approach addresses a different practical question: **Can we identify players likely to establish themselves as NFL starters?** It first standardizes athletic features by position, uses K-means to discover four

player profiles, then evaluates logistic regression and random forest classifiers for the three-plus-year starter outcome. Training on 2000–2015 and testing on 2016–2019 more closely resembles the task of evaluating an upcoming draft class than a random historical split.

The group profiles differed in both starter rates and draft treatment: explosive athletes (40% starters; 89% drafted; average pick 90), big and strong athletes (32%; 80%; pick 109), light and quick athletes (27%; 78%; pick 115), and slow movers (18%; 58%; pick 145). Pro Bowl rates were respectively **20%, 10%, 11%, and 5%**. The model therefore finds meaningful descriptive associations, but draft status and position-related opportunities may partly explain those differences.

On the held-out 2016–2019 classes, **Combine-only logistic regression AUC was 0.654**, increasing to **0.861** with draft pick; **Combine-only random forest AUC was approximately 0.68**, increasing to **0.869** with draft pick. Our results show that at a **0.30 classification cutoff**, the forest identifies **60% of future starters**. This recall value is useful but incomplete without the associated false-positive rate and precision. Our regression results also indicate that a one-standard-deviation improvement in 40-yard speed was associated with approximately **1.59 times the odds** of becoming a starter, and a one-standard-deviation increase in weight with approximately **1.43 times the odds**, conditional on the fitted logistic model. These are associations, not causal effects.



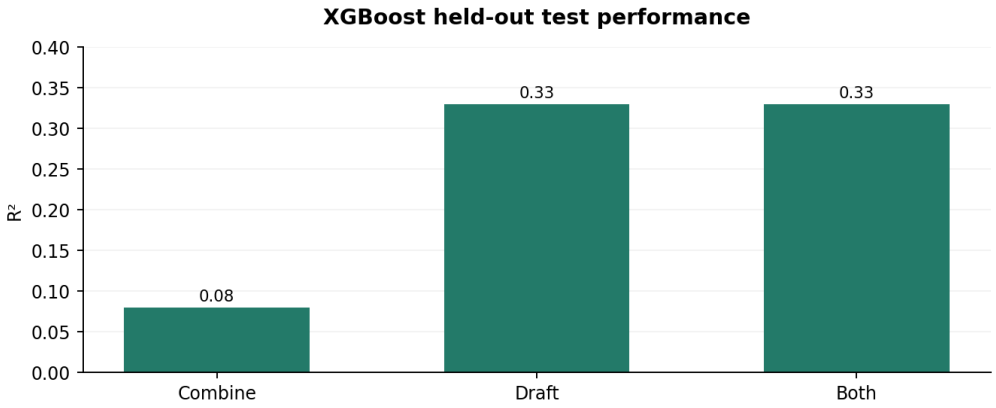
Starter model ROC AUC

Figure 4. Approximate random forest test AUC for starter classification; combine-only value rounded to 0.68.

Does it work? To a degree. In statistical terms, the Combine-only model ranks starter candidates better than chance (AUC ≈ 0.68). The combined model performs substantially better (AUC ≈ 0.87). However, our current results do not include a draft-only AUC for this exact test set, so we **cannot yet claim** that the Combine improves classification beyond draft position. Establishing that comparison is a priority for the final project.

9. Model 3: XGBoost regression

The third approach tests whether a more flexible boosted-tree algorithm improves predictions of career-value percentile. XGBoost was tuned using five-fold cross-validation and evaluated on a held-out test set, with R^2, RMSE, and MAE designated as evaluation measures. Our available results include the held-out R^2 comparisons but **do not provide numerical XGBoost RMSE or MAE values**, so those metrics cannot be reported here without the underlying model outputs.



XGBoost test R squared

Figure 5. Rounded XGBoost held-out R² values for XGBoost.

The Combine-only XGBoost model achieved approximately **$R^2 = 0.08$**, compared with **0.33** using draft information and **0.33** using both. This slightly exceeds the random forest's reported performance, but the difference is small and not accompanied by a statistical uncertainty estimate. A robustness check restricted to the 2000–2019 classes yielded **$R^2 = 0.065$** (Combine), **0.317** (draft), and **0.317** (both). That check supports the central conclusion even when recent, incompletely observed careers are excluded. Position-specific results again suggested modest speed or agility relationships, particularly for running backs, offensive linemen, and tight ends, but no evidence that these small effects consistently translate into major overall gains.

10. Comparing the three approaches and judging success

Approach	Outcome	Combine-only	Draft-based / combined	Main interpretation
Model 1: random forest	Career-value percentile	CV R^2 0.05; test R^2 0.08	Draft and both: test R^2 0.32	Draft dominates; drills add little
Model 2: random forest classifier	Starts 3+ seasons	Test AUC ≈ 0.68	Combine + draft: AUC ≈ 0.87	Predicts above chance; missing draft-only baseline
Model 3: XGBoost	Career-value percentile	Test R^2 ≈ 0.08	Draft and both: test R^2 ≈ 0.33	Slight improvement over forest; no clear incremental Combine gain

These results support three conclusions. **First**, athletic testing does contain information about future NFL outcomes; the starter-classification result is well above chance and some position-specific drill relationships are observable. **Second**, the Combine alone explains only a small fraction of career-value differences; roughly 92% of variation remains unexplained by the Combine-only XGBoost model under the reported test evaluation. **Third**, draft position is a stronger summary predictor, likely because it reflects broader information teams already incorporate, such as college production, film, medical evaluation, and perceived role. That interpretation is plausible but was not independently tested by isolating each of those inputs.

Who can use this now? A scouting department could use the findings to temper reactions to extreme workouts, compare athletic performance within position, and flag profiles for additional review. A front office should **not** use the models as a standalone draft board, claim that explosive training causes better careers, or interpret the 0.33 R^2 as proof of financially profitable draft choices. The practical decision rule remains unvalidated: we have not yet shown that these predictions identify players who outperform their draft slot.

11. Problems encountered and how we addressed them

Unequal career windows. Players drafted in 2024 or 2025 have not had time to accumulate long-term career value. Models 1 and 3 used draft-class percentiles and robustness checks through 2019; Model 2 excluded post-2019 players. These steps reduce but do not eliminate censoring concerns.

Missing Combine tests. Skipped tests were common, especially the shuttle, three-cone, and bench press. Different analyses used missing indicators, position-level filling, or available-case comparisons. The remaining risk is that test participation is selective.

Overfitting and comparison fairness. A fully grown decision tree achieved perfect training fit but negative test R^2. Depth restrictions, pruning, ensembles, and cross-validation improved generalization. Separate hyperparameter tuning for different input groups corrected misleading early estimates of added Combine value.

Data linkage and leakage. Source records included many position labels and career statistics that could reveal outcomes. Position labels were consolidated, PFR identifiers were used to join data, and outcome-related career statistics were excluded from model inputs. Undrafted players were not consistently available for the career-value regression target.

Weak effects and small subgroups. Position-level results varied by era, and the athletic clusters had a low silhouette score. We therefore interpret subgroup cutoffs and cluster labels as exploratory leads rather than stable scouting rules.

12. Future work

First, make evaluation directly comparable. Run all appropriate models on a common historical holdout where possible, document identical preprocessing, report uncertainty intervals, and include RMSE and MAE for XGBoost plus precision, recall, F1, and precision-recall AUC for starter classification. Model 2 especially needs a **draft-only baseline** on the same 2016–2019 test players before incremental value can be claimed.

Second, target draft value rather than raw career value. A potentially more useful question is whether Combine features help predict which players exceed the career value expected at their draft position. This would more directly support finding steals or avoiding busts, while requiring careful treatment of draft position and the fact that undrafted players have different data availability.

Third, test whether the apparent thresholds are real. Use repeated cross-validation or bootstrap samples to assess whether the proposed RB three-cone and TE 40 cutoffs recur in different years and samples. Examine whether missing-drill indicators themselves carry predictive information, and test sensitivity to position grouping and career follow-up length.

Finally, test whether this helps teams make decisions. Before calling the approach ready for NFL decision-making, compare a draft-only ranking with a ranking augmented by Combine predictions on future draft classes. Measure whether the augmented method consistently improves identification of high-value players at

similar draft costs. Without that improvement, the Combine remains useful context rather than a demonstrated source of extra draft value.

13. Conclusion

Our three modeling approaches tell a consistent story. The NFL Combine is **indicative of certain future outcomes**, but not strongly enough to explain most variation in career success by itself. Athletic profiles and speed-related measures are associated with starter likelihood and may matter differently by position. Nevertheless, draft position provides substantially stronger predictions of career value, and adding Combine results did not materially improve the best reported regression results. Our progress so far is in quantifying both the limited signal in athletic testing and its practical limitations. The next step is to test whether the Combine can identify players who will outperform the expectations already embedded in draft position.

References and source notes

Data: nflverse NFL Combine and draft datasets; player linkage through Pro Football Reference identifiers; career-value target derived from PFR weighted approximate value. Repository-specific dataset and code URLs should be inserted when provided by the group.

Figure notes: Figures 1–5 visualize aggregate results from our analyses. They were not regenerated from individual player records for this report.

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1. Kuzmits & Adams (2008).↩

2. Robbins (2010).↩

3. Teramoto et al. (2016).↩